

Scrambled When Wrong, Left Alone When Right: A Prediction-Free Account of Learning in DishBrain

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Abstract

Neurons in a dish learn to play Pong—but the published feedback does not show that they do it by minimizing surprise. In DishBrain the stimulus after a miss is not only unpredictable: it has twice the amplitude of the stimulus after a hit and is followed by eight seconds without play. We prove that such an asymmetry teaches by itself. If a failure perturbs a network’s setting by a random kick of variance σ_m^2 and a success by σ_h^2 , the setting settles with density $\propto 1/[\kappa + (1 - \kappa)P]$, where $\kappa = \sigma_h^2/\sigma_m^2$ and P is the miss rate: fewer misses than chance for $\kappa < 1$, no change at $\kappa = 1$, more misses for $\kappa > 1$. An imaginary-time Schrödinger and a Boltzmann form of this dynamics show that the perturbation strength sets the learning rate ($\propto \sigma_m^2$) but not the level reached, and that perturbations growing with the miss rate would raise that level. The published gameplay data fit this: only the Stimulus condition improves within a session (+0.152 hits per rally, $p = 0.0002$; Silent -0.03 and No feedback -0.002 , not significant), all three conditions end at the same level ($p = 0.22$), and comparisons with Rest are confounded by unequal Rest baselines ($p = 7 \times 10^{-5}$). A spiking network on the chip’s geometry, calibrated to the rates, irregularity, synchrony and absence of bursts in the human cultures’ Rest recordings, carries no sensory-to-motor path at any density from 375 to 1.2×10^5 neurons per mm^2 (10^6 neurons): pulses only suppress the motor regions, with a weak up–down bias that would steer the paddle away from the ball, so it cannot show closed-loop learning. Under local plasticity without reward, its emergent κ is 0.6–0.8 when 75 mV pulses recruit near threshold and above 1 when they recruit widely, and it comes from the eight-second time-out after a miss, not from the pulse amplitude; the network therefore also predicts learning in Silent, which the data do not show. An equal time-out after hits and misses, with stimulation only after misses, decides between time and amplitude.

1 Introduction

In the DishBrain experiment (Kagan et al., 2022), spikes recorded in four motor regions of a cortical culture on a high-density multi-electrode array move a paddle (two regions move it up and two down, arranged in counterbalanced pairs), and the position of the ball is delivered to eight sensory electrodes as a place and rate code. After a hit the culture receives

a “predictable” stimulus: 100 Hz for 100 ms on all eight electrodes at the sensory amplitude, 75 mV. After a miss it receives an “unpredictable” one: 150 mV at 5 Hz at random sites and times for 4 s, followed by 4 s without stimulation. Cultures in this condition improved within a session, and the improvement was interpreted as networks acting to make their input predictable, in line with the free-energy principle (Friston, 2010; Kagan et al., 2022).

The higher voltage after a miss was chosen so that it would force action potentials “regardless of the state the cell was in, thereby being even more disruptive to the culture” (Kagan et al., 2022). The feedback after a miss therefore differs from that after a hit in two ways at once: it is less predictable and more disruptive. The published controls do not separate the two. *No feedback* removes both differences; *Silent* replaces both stimuli by the same interval without stimulation, which still withdraws the sensory input for eight seconds after a miss and hardly at all after a hit. Published criticism of the work concerned mainly the terms “intelligence” and “sentience” (Balci et al., 2023; Kagan et al., 2023).

Here we ask which property of the feedback teaches. We show that the difference in disruption is sufficient (section 2), confirm it in a closed-loop model (section 3), and propose experiments that decide between the accounts (section 5).

2 Learning by Outcome-Contingent Perturbation

Let $\psi \in \Omega$ denote the network’s setting, the state variables that determine its play, and $P(\psi)$ its probability of missing a ball. Suppose that after a miss ψ jumps according to a symmetric kernel, $K(\psi, \psi') = K(\psi', \psi)$, and that after a hit it stays where it is. Nothing tells the network in which direction to change; we call this teacher a *scrambler*.

Proposition 1. *Let Ω be finite, $P > 0$ on Ω and K irreducible. The stationary distribution of the setting over ball approaches is*

$$\rho(\psi) \propto \frac{1}{P(\psi)}, \quad (2.1)$$

and the long-run miss rate is the harmonic mean of P over Ω . It is at most the arithmetic mean, the miss rate of a setting drawn uniformly at random, with equality only if P is constant.

A setting that misses ten times less often is kept ten times longer: good settings are not sought, they are left alone. In general both outcomes perturb the network, with variance σ_m^2 per event after a miss and σ_h^2 after a hit. For small isotropic perturbations the setting diffuses with the state-dependent coefficient

$$D(\psi) = \frac{1}{2} \left[\sigma_m^2 P(\psi) + \sigma_h^2 (1 - P(\psi)) \right]. \quad (2.2)$$

Proposition 2. *Let $\kappa = \sigma_h^2 / \sigma_m^2$ and let the setting diffuse on a bounded domain with reflecting boundary and coefficient (2.2), the step size being set at the point of departure. The stationary density is*

$$\rho(\psi) \propto \frac{1}{\kappa + (1 - \kappa)P(\psi)}. \quad (2.3)$$

The stationary miss rate is below the uniform average of P for $\kappa < 1$ (learning), equal to it for $\kappa = 1$, and above it for $\kappa > 1$ (anti-learning), unless P is constant.

Proofs are in appendix A. The mathematics is classical. For two settings and a kernel that swaps them, Proposition 1 is the “stay with a winner, switch on a loser” rule for the two-armed bandit (Robbins, 1952), and the depth-one automaton with linear tactics, whose behavior is expedient—better than random choice (Tsetlin, 1961, 1963). In general it is the invariant measure of a continuous-time Markov chain whose jump chain is symmetric and whose holding rate is P (Norris, 1997). The stationary density $1/D$ of Proposition 2 is that of hopping with jump rates set at the departure point (van Kampen, 1988; Schnitzer, 1993), and the sign rule is Chebyshev’s sum inequality (Chebyshev, 1882). What is new is the observation that DishBrain’s feedback implements this rule and the experiments that follow from it. Predictability does not enter either statement; what matters is how far each outcome moves the network. The size of the effect is set by the spread of P over the reachable settings; a task in which no setting is much better than the others cannot be learned this way.

The principle has a long history. Ashby’s ultrastable homeostat changes its parameters at random until its essential variables stay within bounds (Ashby, 1952). Woodlice gather in damp places by moving more in dry air (Gunn, 1937), and random walkers accumulate where they move slowly (Schnitzer, 1993); the scrambler is this orthokinesis in the space of network settings. Cortical cultures learn a target response when stimulation is withdrawn as soon as the response occurs, which led to the stimulus regulation principle (Shahaf & Marom, 2001; Marom & Shahaf, 2002), and spiking networks with spike-timing-dependent plasticity learn behaviors that stop external stimulation (Sinapayen et al., 2017; Masumori et al., 2020). None of these works analyzes DishBrain.

2.1 A Schrödinger and a Boltzmann Picture

The diffusion of Proposition 2, $\partial_t \rho = \nabla^2(D\rho)$, becomes self-adjoint after the substitution $\rho = D^{-1/2}\phi$ (Risken, 1989):

$$-\partial_t \phi = H\phi, \quad H = -\sqrt{D} \nabla^2 \sqrt{D} \geq 0. \quad (2.4)$$

With reflecting boundaries the ground state is $\phi_0 \propto D^{-1/2}$, with $E_0 = 0$, and the normalized ground state gives the learned distribution, $\rho_\infty = \hat{\phi}_0^2 \propto 1/D$. In one dimension the change of variable $y = \int d\psi/\sqrt{D}$ brings H to the standard form $-\partial_y^2 + V(y)$. On a finite set of settings $H = \text{diag}(\sqrt{P})(I - K)\text{diag}(\sqrt{P})$, a tight-binding Hamiltonian whose ground state $\phi_0 \propto P^{-1/2}$ reproduces Proposition 1. The evolution is in imaginary time and reversible; nothing is quantum.

Two consequences follow (appendix B). First, the lowest excited level E_1 sets the rate of learning: the distance to the learned distribution decays as $e^{-E_1 t}$. When a hit perturbs nothing ($\sigma_h = 0$), $D = \frac{1}{2}\sigma_m^2 P$ and $H = \sigma_m^2 H_1$ with H_1 fixed by P alone, so the strength of the post-miss perturbation sets how fast a culture learns, $E_1 \propto \sigma_m^2$, but not where it ends. This is a prediction for the published conditions: a weaker disruption learns more slowly toward the same level.

Second, the learned distribution is a Boltzmann distribution,

$$\rho_\infty \propto e^{-U/T}, \quad U = \ln D = \ln[\kappa + (1 - \kappa)P] + \text{const}, \quad T = 1, \quad (2.5)$$

with $U = \ln P$ for $\kappa = 0$, and a true equilibrium, since detailed balance holds. There is no force: the potential is induced by the state dependence of the noise. The temperature is

pinned at 1, because rescaling all perturbations shifts U by a constant; this is the separation of speed and endpoint again. And concentration is weak: occupancies stand in the ratio of the miss rates, not exponentially in them, which is why the gain is only the gap between the arithmetic and harmonic means. The gain can be raised by cooling: if the perturbation per approach grows as P^α , then $\rho_\infty \propto P^{-\alpha}$, an effective temperature $1/\alpha$. A protocol that scales the post-miss stimulation with a culture’s recent miss rate tests this. Finally, like any Markov process, this one lowers the free energy $F[\rho] = \langle U \rangle_\rho - S[\rho]$, the relative entropy to ρ_∞ up to a constant. Here U is the logarithm of the failure rate, not a sensory surprise, so a decrease of free energy during learning does not by itself indicate predictive processing.

3 A Closed-Loop Model

We first checked Proposition 1 on a two-parameter paddle, $p = \text{clip}(ay + b)$ for a ball arriving at height $y \sim U(0, 1)$, with a miss moving (a, b) to a random neighbor on a 31×21 grid. The predicted stationary miss rate is 0.815 and the simulated rate 0.810, against 0.845 for a uniformly random setting.

We then built a DishBrain-like loop (appendix C), first in the abstract form of Proposition 2, with the network’s setting reduced to its sensorimotor weights. The ball’s height relative to the paddle is place-coded on 8 sensory channels and its distance to the paddle wall rate-coded from 4 to 40, as in the experiment. Two motor populations read the active channel through 16 plastic weights in $[-1, 1]$, and their rectified noisy rates move the paddle. The only teacher is a random kick to all weights after each approach, of standard deviation σ_m after a miss and σ_h after a hit, on top of a small spontaneous drift. Sessions of 2000 approaches compare the first quarter with the remaining three, as the experiment compares the first 5 of 20 minutes.

The Silent condition removes the stimulation after a miss but not the eight-second withdrawal of sensory input. In cultures that burst, unstimulated periods let global bursts return (Wagenaar et al., 2005), and bursts make synaptic weights drift (Chao et al., 2005); the curve labelled Silent in Figure 1 shows this burst-driven case as a weaker scrambler ($\sigma_m = 0.03$ against 0.08). It is a conditional prediction, not a reproduction of the data: the Silent cultures of Kagan et al. (2022) were all human-derived, and human cultures in their Rest recordings do not burst (section 3.1); correspondingly, Silent shows no within-session gain in the published data (section 4). In this abstract model a miss is followed by a new serve also in the No-feedback condition, whereas in the experiment No feedback is one continuous rally; the network model below plays the exact protocol.

Over 400 simulated cultures the within-session change in hit fraction (mean $\pm$ s.e.) is $+0.161 \pm 0.014$ for Stimulus, $+0.068 \pm 0.012$ for Silent and $+0.017 \pm 0.007$ for No feedback (Figure 1a). This is the reported ordering, with Silent above No feedback but with a smaller effect (Kagan et al., 2022). The same kick after every outcome gives -0.010 ± 0.007 , and reversed disruption -0.019 ± 0.006 . Performance falls monotonically with κ and crosses the information-free level at $\kappa = 1$ (Figure 1b), as Proposition 2 requires. The model is not fitted to data: the time scale of learning depends on the kick size and the number of plastic parameters, both unknown for a culture. The claim is qualitative, that the contingency in DishBrain’s feedback suffices to produce its effects. The time scale matters, however: in the published sessions a rally lasts $12.7 + 5.9h$ s with h hits (fit to the gameplay data), so a

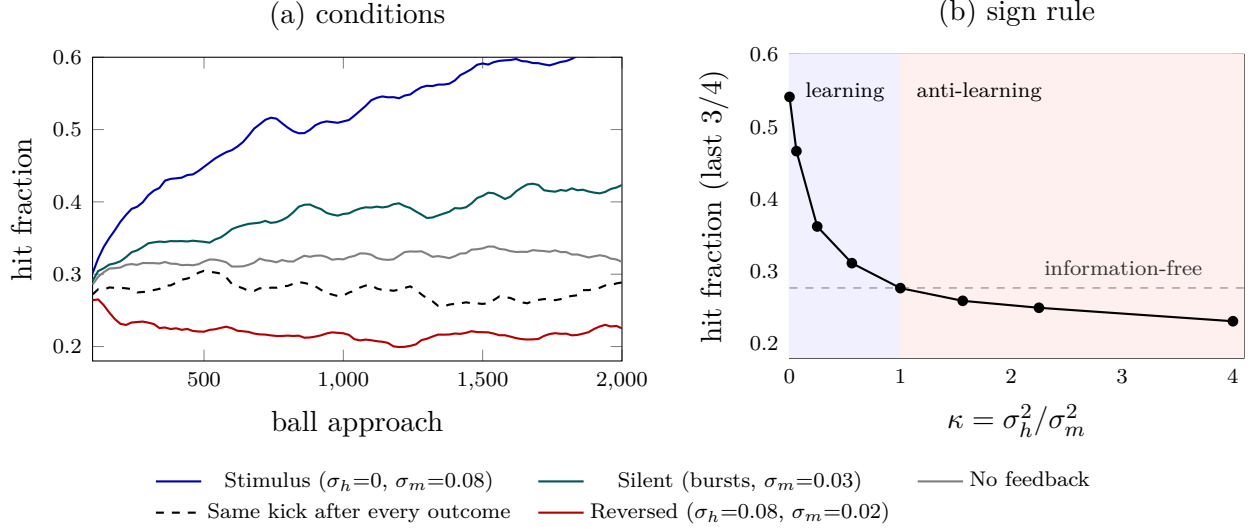


Figure 1: Learning by outcome-contingent perturbation in a closed-loop Pong model (400 simulated cultures). (a) Hit fraction along a session, in a sliding window of 100 approaches. (b) Hit fraction over the last three quarters of a session as a function of κ , with $\sigma_m = 0.08$; the dashed line is the information-free level, $\kappa = 1$. Nothing in the model computes a prediction, a reward or the sign of an error.

20-minute session holds about 170 approaches and 95 misses, against 2000 approaches here. A culture learning within one session therefore needs per-miss perturbations roughly ten times larger than in this model.

3.1 A Network with the Culture’s Geometry

We then asked whether DishBrain’s actual stimuli, acting through local plasticity, realize $\kappa < 1$. A sheet of 3000 leaky integrate-and-fire neurons (80% excitatory, 100 distance-weighted inputs each, short-term depression) covers the 3.85×2.10 mm MaxOne array, and the protocol is replicated exactly: electrode layout, place and rate code, hit and miss programs, the four motor regions with the 20 Hz gain normalization, blanking after stimulation, No feedback as one continuous rally, and session timing (appendix D). The paddle half-length is set so that Rest sessions give the published 0.73 hits per rally.

Calibration. Free parameters were fitted to eight statistics of the 168 Rest recordings of the human cultures (Kagan et al., 2022), computed identically for model and data (Table 1). The fit is within the data’s interquartile range except for rates about 0.1 Hz high and pairwise correlations near zero. The Rest statistics fix rates, irregularity, synchrony and the absence of bursts, but not recurrent coupling: parameter sets with excitatory weights spanning a factor of 40 fit about equally well, so we carry three (weak, medium, strong coupling).

No sensorimotor channel at any density we can simulate. Single pulses at the stimulation electrodes produced no detectable change in the motor regions, about 1 mm away, in any set or recruitment assumption (fraction of cases with $z > 3$: 0.00; with medium coupling, 150 mV pulses slightly suppress all regions alike). Consequently the paddle is not driven by the ball: in 20-minute sessions (medium coupling, gated rule, 32 cultures) hits per rally

changed by -0.11 in Stimulus, $+0.09$ in Silent and $+0.04$ in No feedback, none significantly (Wilcoxon $p = 0.12, 0.07, 0.49$). At 3000 neurons the model has about 375 neurons per mm^2 , far fewer than a culture of 10^6 plated cells, and the real cultures respond across the whole array during stimulation (Kagan et al., 2022, Fig. 4C). We therefore repeated the test at 10^5 , 3×10^5 and 10^6 neurons ($1.2\text{--}12 \times 10^4$ per mm^2) with the same wiring rule, after re-scoring the best parameter sets at each size against the Rest statistics (appendix D). Density does not remove the coupling degeneracy: at 10^6 neurons the best weak, medium and strong sets have losses 0.71, 1.25 and 1.07. Nor does it create a path. Motor regions were excited in at most 3% of cases at 10^5 and in none at larger sizes. What grows with density is suppression: 150 mV pulses lower motor counts by up to 13%, with $z < -3$ in up to 61, 80 and 97% of cases at the three sizes, alike in up and down regions. The up-minus-down difference exceeds $|z| = 3$ in at most 6% of cases at 10^5 and 3×10^5 and 25% at 10^6 . Its pattern across the eight electrodes, which are ordered by ball height relative to the paddle, is anticorrelated with correct steering ($r = -0.72, -0.88$ and -0.92 for 150 mV at the three sizes) and similar between sizes although the topologies are independent ($r = 0.47\text{--}0.76$): a small, reproducible bias that pushes the paddle away from the ball. The 500-ms, 40 Hz place-code trains of play give the same result: the up-minus-down change exceeds $|z| = 3$ in 3.5, 6.2 and 11% of cases on average at the three sizes (at most 9, 19 and 44%), its pattern across electrodes anticorrelates with correct steering ($r = -0.96, -0.95$ and -0.90) and is nearly the same at all sizes ($r = 0.92\text{--}0.95$), and although the bias grows with density it stays about 0.2% of the motor counts on average (at most 1%). Closed-loop learning therefore cannot be tested in this model at any density we can simulate. What is missing is not cell count but a focused path over about 1 mm, which random distance-dependent wiring with 100 inputs per neuron does not provide. What can be measured is the quantity Proposition 2 depends on.

Emergent κ . We logged the total squared change of the excitatory weights after every hit and every miss, under two local rules without reward or error signal: activity-gated random updates and spike-timing-dependent plasticity (Table 2). When a 75 mV pulse recruits few cells, or many cells only up to threshold, as the authors intended in choosing 75 mV “to prevent forcing hyperpolarised cells from firing” (Kagan et al., 2022), κ per approach is 0.6–0.8 in all sets and under both rules: the learning regime. It is the same in Stimulus, in Predictable (protocol B) and in Silent, because it comes from time, not stimulation: a miss is followed by eight seconds without play and a new serve, a hit by the return of the ball, and plasticity accrues at a nearly constant rate. With small recruitment the 150 mV random program adds only 15% to the displacement of eight seconds of silence. When 75 mV pulses recruit as widely as 150 mV pulses, the 80-pulse hit program dominates and $\kappa > 1$: anti-learning.

Under local plasticity the network therefore predicts learning for Stimulus and Silent alike and none for No feedback. The data show learning in Stimulus only. Either the 150 mV stimulation perturbs synapses in real cultures far more than activity-gated rules do in a sparse model, or Silent’s missing gain reflects its high starting level (0.849 hits per rally at T1, above Stimulus’s end point). The model leaves the amplitude account untested rather than refuted; protocol F (Table 3) separates the two.

Statistic (Rest, human cultures)	Data: median [IQR]	Model: median [IQR]
rate per active unit, median (Hz)	0.30 [0.26, 0.39]	0.42 [0.38, 0.45]
rate per active unit, mean (Hz)	0.55 [0.46, 0.69]	0.72 [0.65, 0.77]
CV of inter-spike intervals	0.97 [0.93, 1.02]	0.96 [0.96, 0.97]
pairwise correlation, 100 ms bins	0.0025 [0.0007, 0.012]	0.0000 [−0.0002, 0.0001]
Fano factor of population count	1.08 [0.92, 1.46]	0.91 [0.89, 0.93]
units co-active in 50 ms, 99th pct.	0.086 [0.078, 0.100]	0.086 [0.086, 0.099]
units co-active in 50 ms, max.	0.16 [0.13, 0.23]	0.14 [0.12, 0.14]
network events per minute	0 [0, 0.15]	0 [0, 0]

Table 1: Calibration of the network to the Rest recordings of the human cultures of [Kagan et al. \(2022\)](#) (data: 168 recordings, 24 chips; model: 64 ten-minute runs of the best set, 81 active units subsampled per run, as in the data).

Rule	75 mV recruitment	κ per approach			programs only	
		Stimulus	Predictable	Silent	κ_{prog}	$\frac{150 \text{ mV miss}}{\text{silent miss}}$
gated	small	0.65	0.65	0.63	0.12	1.16
gated	wide, near-threshold	0.72	0.72	0.82	0.35	2.71
gated	wide	1.26	1.28	1.11	0.92	2.83
gated	widest	2.95	2.93	1.60	4.60	4.03
STDP	small	0.63	0.63	0.65	0.10	1.15
STDP	wide	1.11	1.10	0.97	0.75	2.55
STDP	widest	2.96	2.94	1.49	4.07	3.21

Table 2: Emergent κ in the calibrated network: squared excitatory-weight change from one outcome to the next after hits, divided by the same after misses (means over the three coupling sets, 8 cultures each, 5-minute sessions; values per set in **network/results**). κ_{prog} : the same for the feedback programs alone. Last column: displacement of the Stimulus miss program relative to eight seconds of silence.

4 Re-analysis of the Published Data

The per-rally gameplay data of the feedback study are public ([Kagan et al., 2022](#); OSF, <https://doi.org/10.17605/osf.io/5u6qv>), and our analysis script is in the folder **reanalysis**. Taking the culture as the unit and hits per rally as the measure, in the first 5 minutes of gameplay (T1) and the remaining 15 (T2):

- Stimulus rises within the session, from 0.687 to 0.839 (+0.152, Wilcoxon $p = 0.0002$, $n = 24$), in every cell line; Silent (−0.033, $p = 0.35$, $n = 17$) and No feedback (−0.002, $p = 1$, $n = 15$) do not.
- At T2 the three conditions do not differ (Kruskal–Wallis $p = 0.22$). Stimulus reaches, within the session, the level the other groups already had.

Both points fit section 2.1: when a hit perturbs little, the strength of the post-miss perturbation sets the speed of learning, not the level reached. Two methods caveats apply to the published

Protocol	After a hit	After a miss	Predictability account	Perturbation account
A. Published	75 mV, 100 Hz, 100 ms	150 mV random, 4 s; 4 s without input	learns	learns
B. Predictable penalty	as A	150 mV, same dose as A, in a fixed sequence	weak or none	learns as A
C. Quenched penalty	as A	8 s of random, rapid, low-amplitude stimulation that suppresses bursts	learns	weak or none
D. Reversed	150 mV fixed pattern, 1 s	75 mV random, 1 s	learns	below E
E. Information-free	same random stimulus	same random stimulus	none	none
F. Equal time-out	8 s pause	150 mV random, 4 s; 4 s pause	learns	learns only if amplitude matters

Table 3: Protocols that separate the accounts. B and C are the minimal pair; D tests the sign rule of Proposition 2; E is the reference level for D; F separates the time-out after a miss from its stimulation (section 3.1).

comparisons, which express gameplay relative to each culture’s Rest session. The Rest baselines differ between the condition groups (0.666, 0.775 and 0.787 hits per rally; $p = 7 \times 10^{-5}$), with the Stimulus cultures worst at rest, and the groups differ in cell type (Silent has no mouse cultures). Finally, as a suggestive signature of a scrambler, the lag-1 correlation of consecutive rally lengths falls from Rest to gameplay in Stimulus ($0.098 \rightarrow 0.026$, $p = 0.0005$) but not in Silent or No feedback; the Stimulus cultures’ high Rest persistence carries the same baseline caveat, so this is consistent with, not evidence for, perturbation after misses.

5 Experiments That Decide

The two accounts agree on every condition tested so far and disagree on those in Table 3. Protocols B and C each change one property of the feedback after a miss while holding the other fixed; protocol D reverses which outcome is the more disruptive.

Two analyses of existing recordings also decide. *Per-event displacement*: measure how far each feedback event moves the network’s state, for example the change in the evoked response to a fixed probe or in functional connectivity across the event. The perturbation account predicts that a miss moves the state further than a hit and that, across cultures and conditions, the within-session gain grows with the difference of the mean squared displacements. *Bursts*

after a miss: in the Silent condition, the number of spontaneous bursts in the eight seconds after each miss should predict a culture’s gain.

6 Discussion

If the perturbation account holds, DishBrain demonstrates a generic property of plastic networks under outcome-contingent perturbation rather than predictive processing in particular. The accounts are not mutually exclusive: a surprise-minimizing system may implement exactly this stabilization, and the present data cannot tell them apart. The account also explains why the improvement did not carry over between sessions (Kagan et al., 2022): a setting held only by the absence of perturbation is diffused by spontaneous bursting between sessions (Chao et al., 2005), which predicts that quenching bursts between sessions should improve retention. For comparisons of sample efficiency between cultures and learning algorithms, the natural baseline is a learner that uses only the hit-or-miss signal by perturbing on failure.

Two limitations apply. Real stimulation does not perturb synapses isotropically, and a patterned 100 Hz burst can itself induce pathway-specific potentiation and depression (Jimbo et al., 1999), so the “predictable” stimulus may push the network in a consistent direction rather than leave it alone; the displacement analysis measures such effects directly. The network model (section 3.1) carries no sensorimotor channel even at 10^6 neurons, so it tests the perturbation contingency, not closed-loop learning; under local rules it attributes $\kappa < 1$ to the time-out after a miss rather than to the stimulation, and does not explain why Silent did not learn. The burst reading of Silent applies only to bursting cultures.

Code Availability

The abstract model (`scrambler.py`, standard-library Python), the network simulator and its calibration (`network/`, PyTorch on a GPU), the re-analysis (`reanalysis/`) and the data behind the figures and tables are in the folder `papers/dishbrain-scrambler` of the repository

<https://github.com/kladkogex/20-watt-computer>.

A Proofs

Proposition 1. Between approaches the setting moves from ψ to ψ' with probability $P(\psi)K(\psi, \psi')$. The measure $\rho \propto 1/P$ satisfies detailed balance, $\rho(\psi)P(\psi)K(\psi, \psi') = \rho(\psi')P(\psi')K(\psi', \psi)$, because K is symmetric; since K is irreducible and $P > 0$, it is the unique stationary distribution. The long-run miss rate is $\sum_{\psi} \rho(\psi)P(\psi) = |\Omega| / \sum_{\psi} 1/P(\psi)$, the harmonic mean of P , which is at most the arithmetic mean by the inequality of the means, with equality only for constant P . (Equivalently, the jump chain K has the uniform invariant measure, and the mean holding time in ψ is $1/P(\psi)$; Norris, 1997.)

Proposition 2. When the size of a step is set at the point of departure, the density obeys the Itô form of the Fokker–Planck equation, $\partial_t \rho = \nabla^2(D\rho)$ (Schnitzer, 1993). With a reflecting boundary the stationary flux vanishes, so $D\rho$ is constant and $\rho \propto 1/D \propto 1/[\kappa + (1 - \kappa)P]$.

Write $w = 1/[\kappa + (1 - \kappa)P]$ and let $\mathbb{E}_u$ denote the average over the uniform measure on the domain, the stationary law of information-free kicks. The stationary miss rate is $\mathbb{E}_u[Pw]/\mathbb{E}_u[w]$. For $\kappa < 1$, w is a decreasing function of P , and Chebyshev's sum (association) inequality (Chebyshev, 1882) gives $\mathbb{E}_u[Pw] \leq \mathbb{E}_u[P] \mathbb{E}_u[w]$, with equality only if P is constant; hence the miss rate is at most $\mathbb{E}_u[P]$. For $\kappa = 1$, w is constant. For $\kappa > 1$, w increases with P and the inequality reverses.

B The Schrödinger and Boltzmann Pictures

Self-adjoint form. Put $\rho = D^{-1/2}\phi$ in $\partial_t \rho = \nabla^2(D\rho)$. Then $D^{-1/2}\partial_t \phi = \nabla^2(\sqrt{D}\phi)$, that is $\partial_t \phi = -H\phi$ with $H = -\sqrt{D}\nabla^2\sqrt{D}$. The flux is $J = -\nabla(D\rho) = -\nabla(\sqrt{D}\phi)$, so a reflecting boundary is the Neumann condition $\partial_n(\sqrt{D}\phi) = 0$. For two such functions Green's identity gives $\langle \chi, H\phi \rangle = \int \nabla(\sqrt{D}\chi) \cdot \nabla(\sqrt{D}\phi)$, which is symmetric and non-negative: H is self-adjoint and $H \geq 0$. $H\phi_0 = 0$ requires $\nabla^2(\sqrt{D}\phi_0) = 0$ with Neumann data, so $\sqrt{D}\phi_0$ is constant and $\phi_0 \propto D^{-1/2}$, $E_0 = 0$. Expanding $\phi(t) = \sum_n c_n e^{-E_n t} \phi_n$ in the eigenfunctions of H , $\rho(t) - \rho_\infty$ decays as $e^{-E_1 t}$. With $\hat{\phi}_0$ normalized, $\rho_\infty = D^{-1/2}\hat{\phi}_0/\|D^{-1/2}\| = \hat{\phi}_0^2$. For $\sigma_h = 0$, $\sqrt{D} = \sigma_m \sqrt{P/2}$ and $H = \sigma_m^2 H_1$ with $H_1 = -\sqrt{P/2}\nabla^2\sqrt{P/2}$: every eigenvalue scales as σ_m^2 , and the eigenfunctions, ρ_∞ among them, do not depend on σ_m .

Standard form in one dimension. With $dy = d\psi/\sqrt{D}$ and $q(y) = \sqrt{D}$, the measure is $d\psi = q dy$; the function $\chi = q^{1/2}\phi$ has the flat norm $\int \chi^2 dy = \int \phi^2 d\psi$, and a direct calculation gives $q^{1/2}Hq^{-1/2} = -\partial_y^2 + V(y)$ with

$$V = \frac{3}{4} \left(\frac{q'}{q} \right)^2 - \frac{1}{2} \frac{q''}{q} = \frac{\chi_0''}{\chi_0}, \quad \chi_0 = q^{-1/2} = D^{-1/4}, \quad (\text{B.1})$$

primes denoting d/dy . The ground state χ_0 has $E_0 = 0$ by construction.

Finite set of settings. Per approach the distribution changes as $\rho_{t+1} = \rho_t - (I - K)\text{diag}(P)\rho_t$ (a miss, with probability P , moves the setting by K , which is symmetric). With $\phi = \text{diag}(\sqrt{P})\rho$ this reads $\phi_{t+1} = \phi_t - H\phi_t$, $H = \text{diag}(\sqrt{P})(I - K)\text{diag}(\sqrt{P})$, which is symmetric and non-negative because $I - K$ is. Irreducibility makes the constants the only null vectors of $I - K$, so $H\phi_0 = 0$ exactly for $\phi_0 \propto P^{-1/2}$ and $\rho_\infty \propto P^{-1}$: Proposition 1. For small steps the per-approach factor $1 - E_1$ is e^{-E_1} to leading order.

Boltzmann form. $\nabla^2(D\rho) = \nabla \cdot [D(\nabla\rho + \rho\nabla U)]$ with $U = \ln D$: gradient dynamics in the potential U with mobility D and temperature 1, although the Itô drift of the underlying process is zero. The stationary flux vanishes everywhere, so $\rho_\infty \propto e^{-U}$ is an equilibrium with detailed balance in any dimension. Rescaling σ_h and σ_m by c multiplies D by c^2 , shifting U by $2\ln c$ and leaving ρ_∞ unchanged. If instead the variance per approach is $D \propto P^\alpha$ (a per-miss variance growing as $P^{\alpha-1}$), then $\rho_\infty \propto P^{-\alpha} = e^{-\ln P/(1/\alpha)}$. For any Markov process with stationary law $\rho_\infty \propto e^{-U}$, $F[\rho] = \int \rho U + \int \rho \ln \rho = \text{KL}(\rho \| \rho_\infty) - \ln Z$ does not increase in time. These statements are checked numerically in `theory_check.py`.

C Simulation Details

Court and ball. The court is 1×1 with the paddle on the right wall; the paddle has half-length 0.1 and its center is confined to $[0.1, 0.9]$. Each serve starts at the far wall with horizontal speed 1 (one crossing per second) and vertical speed uniform in $[-0.8, 0.8]$; the ball reflects from the other walls. A hit ($|y - p| < 0.1$ at the paddle wall) returns the ball; a miss is followed by a new serve. The loop runs in steps of 20 ms (DishBrain uses 10 ms).

Sensory code and readout. The ball’s height relative to the paddle, clipped to $[-0.5, 0.5]$, selects one of 8 channels, $k = \lfloor 8(y - p + 0.5) \rfloor$, and its horizontal position x sets the rate $s = (4 + 36x)/40$, from 0.1 at the far wall to 1 at the paddle. The motor rates are $r_{\uparrow} = [W_{\uparrow}[k]s + \xi_{\uparrow}]_+$ and $r_{\downarrow} = [W_{\downarrow}[k]s + \xi_{\downarrow}]_+$ with independent Gaussian noise of standard deviation 0.35 per step, and the paddle moves by $1.5(r_{\uparrow} - r_{\downarrow})$ court heights per second.

Plasticity. The 16 weights start uniformly random in $[-1, 1]$. After every approach each weight receives a Gaussian kick of standard deviation $\sigma_0 = 0.01$, and then one of standard deviation σ_h after a hit or σ_m after a miss; weights reflect at ± 1 , so information-free kicks leave the uniform law invariant. Conditions: Stimulus $(\sigma_h, \sigma_m) = (0, 0.08)$; Silent $(0, 0.03)$; No feedback $(0, 0)$; same kick after every outcome $(0.08, 0.08)$; reversed disruption $(0.08, 0.02)$. For Figure 1b, $\sigma_m = 0.08$ and $\sigma_h \in \{0, 0.02, 0.04, 0.06, 0.08, 0.10, 0.12, 0.16\}$.

Sessions and statistics. Each of 400 simulated cultures plays one session of 2000 approaches; the hit fraction over the first 500 approaches is compared with that over the remaining 1500.

Grid check of Proposition 1. $a \in \{-1, -0.9, \dots, 2\}$, $b \in \{-1, -0.9, \dots, 1\}$; $P(a, b)$ is evaluated on 400 equally spaced ball heights. After a miss the setting moves to a uniformly chosen neighbor; moves off the grid or into one of the seven never-miss settings are rejected, which keeps the kernel symmetric on the remaining states. The chain is run for 3×10^6 approaches.

D The Network Model

Network. $N = 3000$ leaky integrate-and-fire neurons ($\tau_m = 20$ ms, threshold 1, reset 0, refractory 2 ms) at uniform random positions on 3.85×2.10 mm; the first 80% excitatory. Each neuron receives $K = 100$ inputs drawn without replacement with probability $\propto e^{-d/\lambda}$, $\lambda = 1$ mm (of the order of the input numbers estimated for dissociated cultures, Soriano et al., 2008; the excitatory fraction and λ are assumptions). Exponential current synapses ($\tau_e = 5$, $\tau_i = 10$ ms); short-term depression of excitatory synapses (release fraction U , recovery 500 ms); Ornstein–Uhlenbeck membrane noise of stationary s.d. σ ; a fixed per-neuron drive $\mu + \mu_{sd} \xi_i$. Integration is exact for the linear parts with $dt = 1$ ms.

Calibration. $\mu, \sigma, \mu_{sd}, J_{EE}, J_{IE}, J_{EI} = J_{II}, U$ and an adaptation strength were drawn at random (weights log-uniformly), 512 sets per round, and refined in two further rounds around the best 16 (60, 90 and 120 s of spontaneous activity). The loss is $\sum_s ((v_s - \text{med}_s)/\text{IQR}_s)^2$ over the eight statistics of Table 1, computed by the same code for data and model after subsampling the model’s active units (≥ 0.1 Hz) to the data’s median of 81. Best set (weak coupling): $\mu = 0.20$, $\sigma = 0.26$, $\mu_{sd} = 0.14$, $J_{EE} = 0.0066$, $J_{IE} = 0.079$, $J_{EI} = 0.066$, $U = 0.089$; medium: $J_{EE} = 0.118$, $J_{EI} = 0.61$; strong: $J_{EE} = 0.153$, $J_{EI} = 0.47$ (losses 0.52, 0.68, 0.77; full values in `network/calib_sets.json`).

Stimulation and protocol. Electrode positions follow Fig. 4D of Kagan et al. (2022): eight stimulation electrodes in a zig-zag across the sensory band and four 31×33 -electrode motor regions, counterbalanced up/down. A 75 mV pulse raises the membrane of neurons within r_{75} of the electrode by k_{75} ; a 150 mV pulse forces a spike within r_{150} . Recruitment assumptions (r_{75}, k_{75}, r_{150}): small (0.05 mm, 1.2, 0.10 mm), wide near-threshold (0.12, 0.7, 0.20), wide (0.12, 1.2, 0.20), widest (0.20, 1.5, 0.23). The decoder sums spikes per region every 10 ms, scales each region by a running (5 s) gain toward 20 Hz, and moves the paddle by 0.01 court heights toward the larger of the up and down sums; readout is blanked for 5 ms after every stimulation command. The ball crosses the court in 2.9 s (fit to the gameplay data); the paddle half-length 0.214 gives the published Rest level of 0.73 hits per rally. A 60 s burn-in without plasticity precedes each session.

Plasticity and displacement. Activity-gated rule: when an excitatory neuron spikes, each excitatory input weight changes by $\pm \eta w_{\max} x_j$ with random sign, x_j the input's 20 ms trace. STDP: additive pair rule with $A_+ = \eta w_{\max}$, $A_- = 1.05 A_+$, 20 ms traces. Weights stay in $[0, w_{\max}]$, $w_{\max} = 2J_{EE}$. The simulator logs the cumulative $\sum \Delta w^2$ at every outcome and at the end of every feedback program; κ per approach is the mean increment from a hit to the next outcome divided by that from a miss.

Large networks. For $N > 2 \times 10^4$ the inputs are sampled without the $N \times N$ distance matrix: each input lies at a Gamma(2, λ)-distributed distance in a uniform direction, which gives the same density $\propto e^{-d/\lambda}$ per area, and is a random neuron of the grid cell (side equal to the mean neuron spacing) containing that point; draws outside the sheet, into an empty cell or onto the neuron itself are repeated; $K = 100$ at all sizes. Re-calibration: the best 8 sets of the 3000-neuron search in each coupling range (weak $J_{EE} < 0.03$, medium 0.05–0.12, strong 0.12–0.3) were re-scored with 120 s of spontaneous activity (two independent cultures per set at 10^5 and 3×10^5 , one at 10^6), and the best per range carried forward (`scale_calib.py`). Single pulses: 30 trials per electrode, counts in the 50 ms after against the 50 ms before a pulse, two cultures (one at 10^6) per set, recruitment and amplitude (`scale_evoked.py`). Place-code trains: 75 mV pulses at 40 Hz for 500 ms at one electrode against the 500 ms before, 20 trials per electrode, four cultures (two at 10^6) (`scale_train.py`). Correct steering is an up-minus-down change increasing with the electrode index, which grows with the ball's height relative to the paddle; `scale_summary.py` computes the numbers quoted in section 3.1.

Implementation and checks. All runs are simulated together on one GPU (NVIDIA GB10): state tensors carry a run axis, synapses propagate only from neurons that spiked, the neuron update is one fused kernel, and each 10 ms block is replayed as a CUDA graph. 288 five-minute sessions take 6.5–8 min. Firing rates, irregularity and correlations agree with a dense NumPy reference of the same equations to within sampling error (`validate.py`). At most $12R$ spikes per millisecond are propagated; in the 288-run batches this cap was exceeded during synchronous stimulation steps, which drops propagation (and plasticity) of some spikes during the hit tetanus, so the hit-program displacement and hence κ are, if anything, underestimated.

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